Appendix E. Manual design of the two building cases

To provide the reference for the comparison in Section 5.3.2, the two building cases were also designed by human designers; the quantitative results are listed in Tables 14 and 15. Human designers typically use Excel spreadsheets for statistical analysis and basic design calculations. First, they determine the length of each PWP wall segment (Length), then divide it by the length of a standard PWP panel (WStdPnl) to calculate the number of standard panels (nStdPnl). The remaining non-standard portion of each segment is defined as the spare length (Lspare) and is then grouped. During grouping, the spare panels are first sorted by height (hsort), from shortest to tallest, and their spare lengths are reordered to follow this height order (Lsort1). Panels of the same height are then sorted by width, from narrowest to widest (Lsort2). Finally, the sorted panels are combined so that the accumulated width of each group does not exceed that of a standard panel (Lsum); each completed group forms one standard panel (ngrp), and summing these within each height gives the number of standard panels assembled from spare panels (nspare). Together with the directly cut standard panels, this yields the total number of standard panels (SumAllStd) and, in turn, the number of PWP bodies (numBody).

Once the numerical design is completed, the designer must manually draft the CAD drawings, determine the positions from the lengths, assign numbers according to position, and match these numbers to the corresponding standard PWP panels and PWP bodies (Section 5.3.2).

Table 14

Quantitative results of the human design for Case 1

Segment coordinates (mm)								Standard-panel layout (mm)				Sorted segments (mm)						Panel summary (mm)			
NO.	**StartX**	**StartY**	**StartZ**	**EndX**	**EndY**	**EndZ**	**Height**	**Length**	**W_{StdPnl}**	**n_{StdPnl}**	**Len_{spare}**	**h_{sort}**	**L_{sort1}**	**L_{sort2}**	**L_{sum}**	**n_{grp}**	**n_{spare}**	**Items**	**2500**	**2600**	**3000**
1	13350	3750	0	16400	3750	0	2600	3050	605	5	25	2400	540	540				numStd1	32	27	18
2	16400	3750	0	16400	5300	0	2500	1550	605	2	340	2500	340	30				numStd2	10	1	7
3	17450	5150	0	18150	5150	0	3000	700	605	1	95	2500	30	30	600	1		**sumAllStd**	**42**	**28**	**25**
4	18150	5150	0	18150	2700	0	2500	2450	605	4	30	2500	300	95				panelAbody	12	12	12
5	13350	6350	0	16400	6350	0	2600	3050	605	5	25	2500	240	185				**numBody**	**4**	**3**	**3**
6	16400	6500	0	16400	6200	0	2500	300	605	0	300	2500	300	240	520	1					
7	27100	5200	0	27100	3750	0	2500	1450	605	2	240	2500	545	250							
8	27100	3750	0	30150	3750	0	2600	3050	605	5	25	2500	250	250	500	1					
9	27100	6500	0	27100	6200	0	2500	300	605	0	300	2500	250	300							
10	27100	6350	0	30150	6350	0	2600	3050	605	5	25	2500	545	300	600	1					
11	19950	8550	0	18800	8550	0	2500	1150	605	1	545	2500	325	315	315	1					
12	19950	8800	0	19950	8550	0	2500	250	605	0	250	2500	500	325	325	1					
13	23550	8800	0	23550	8550	0	2500	250	605	0	250	2500	95	340	340	1					
14	23550	8550	0	24700	8550	0	2500	1150	605	1	545	2500	30	500	500	1					
15	19950	11450	0	19950	10400	0	3000	1050	605	1	445	2500	185	545	545	1					
16	23550	11450	0	23550	10400	0	3000	1050	605	1	445	2500	315	545	545	1	10				
17	3700	20400	0	4100	20400	0	2600	400	605	0	400	2600	25	25							
18	5000	20400	0	7100	20400	0	3000	2100	605	3	285	2600	25	25							
19	27950	21700	0	27950	25050	0	2500	3350	605	5	325	2600	25	25							
20	29050	20350	0	29050	19950	0	3000	400	605	0	400	2600	25	25							
21	29050	19750	0	29050	19250	0	2500	500	605	0	500	2600	400	140							

22	29050	17650	0	29050	16950	0	2500	700	605	1	95	2600	140	225	465	1	
23	34100	22850	0	34100	21500	0	2600	1350	605	2	140	2600	225	400	400		1
24	33500	20300	0	35800	20300	0	3000	2300	605	3	485	3000	95	95			
25	36900	20300	0	36700	20300	0	3000	200	605	0	200	3000	445	95			
26	36900	20300	0	36900	23300	0	3000	3000	605	4	580	3000	445	200	390	1	
27	35150	21500	0	36900	21500	0	2400	1750	605	2	540	3000	285	280			
28	25350	2700	0	25350	5150	0	2500	2450	605	4	30	3000	400	565	565	1	
29	25350	5150	0	26050	5150	0	3000	700	605	1	95	3000	485	400	400	1	
30	2700	23100	0	2700	20400	0	3000	2700	605	4	280	3000	200	445	445	1	
31	2700	21600	0	4700	21600	0	2500	2000	605	3	185	3000	580	445	445	1	
32	10200	24850	0	10200	21600	0	2600	3250	605	5	225	3000	95	485	485	1	
33	13100	21600	0	8550	21600	0	2500	4550	605	7	315	3000	280	580	580	1	7

Table 15

Quantitative results of the human design for Case 2

Segment coordinates (mm)								Standard-panel layout (mm)				Sorted segments (mm)						Panel summary (mm)			
NO.	**StartX**	**StartY**	**StartZ**	**EndX**	**EndY**	**EndZ**	**Height**	**Length**	**W_{StdPnl}**	**n_{StdPnl}**	**Len_{spare}**	**h_{sort}**	**L_{sort1}**	**L_{sort2}**	**L_{sum}**	**n_{grp}**	**n_{spare}**	**Items**	**2500**	**2600**	**3000**
1	15800	30400	0	15800	31850	0	2500	1450	605	2	240	2500	240	190				numStd1	20	12	15
2	15700	30200	0	15700	30400	0	2500	200	605	0	200	2500	200	190				numStd2	12	2	8
3	15800	26600	0	18750	26600	0	2600	2950	605	4	530	2500	440	190	570	1		**sumAllStd**	**32**	**14**	**23**
4	15800	26600	0	15800	24950	0	2500	1650	605	2	440	2500	190	200				panelAbody	12	12	12
5	15800	26600	0	15800	28000	0	2500	1400	605	2	190	2500	240	200				**numBody**	**3**	**2**	**2**
6	14200	20350	0	14200	22300	0	3000	1950	605	3	135	2500	200	200	600	1					
7	14200	22300	0	14500	22300	0	3000	300	605	0	300	2500	440	240							
8	14500	22300	0	14500	23500	0	3000	1200	605	1	595	2500	190	240	480	1					
9	3200	14100	0	1750	14100	0	2500	1450	605	2	240	2500	200	240							
10	3400	14000	0	3200	14000	0	2500	200	605	0	200	2500	440	250	490	1					
11	7000	14100	0	7000	17050	0	2600	2950	605	4	530	2500	190	250							
12	7000	14100	0	8650	14100	0	2500	1650	605	2	440	2500	445	250	500	1					
13	7000	14100	0	5600	14100	0	2500	1400	605	2	190	2500	250	250							
14	13250	12500	0	11300	12500	0	3000	1950	605	3	135	2500	250	250	500	1					
15	11300	12500	0	11300	12800	0	3000	300	605	0	300	2500	250	250	250	1					
16	11300	12800	0	10100	12800	0	3000	1200	605	1	595	2500	445	440	440	1					
17	30700	20100	0	30500	20100	0	2500	200	605	0	200	2500	250	440	440	1					
18	26900	23150	0	26900	20200	0	2600	2950	605	4	530	2500	250	440	440	1					
19	26900	20200	0	25250	20200	0	2500	1650	605	2	440	2500	250	445	445	1					
20	26900	20200	0	28300	20200	0	2500	1400	605	2	190	2500	240	445	445	1	12				
21	22600	18600	0	20650	18600	0	3000	1950	605	3	135	2600	530	530	530	1					

22	22600	18600	0	22600	18950	0	3000	350	605	0	350	2600	530	530	530	1	
23	22600	18950	0	23850	18950	0	3000	1250	605	2	40	2600	530	530			2
24	16900	8300	0	16900	9350	0	3000	1050	605	1	445	3000	135	40	570	1	
25	16900	6450	0	15850	6450	0	2500	1050	605	1	445	3000	300	135			
26	16900	6700	0	16900	6450	0	2500	250	605	0	250	3000	595	135			
27	15350	4200	0	15100	4200	0	2500	250	605	0	250	3000	135	135	405	1	
28	15100	2850	0	15350	2850	0	2500	250	605	0	250	3000	300	300			
29	20200	8300	0	20200	9350	0	3000	1050	605	1	445	3000	595	300	600	1	
30	21250	6450	0	20200	6450	0	2500	1050	605	1	445	3000	135	350	350	1	
31	20200	6450	0	20200	6700	0	2500	250	605	0	250	3000	350	445	445	1	
32	22000	4200	0	21750	4200	0	2500	250	605	0	250	3000	40	445	445	1	
33	21750	2850	0	22000	2850	0	2500	250	605	0	250	3000	445	595	595	1	
34	32150	20224	0	30700	20224	0	2500	1450	605	2	240	3000	445	595	595	1	8